SURGSTREAM: A WORKFLOW-ANCHORED PROACTIVE STREAMING FRAMEWORK AND BENCHMARK FOR REAL-TIME SURGICAL VIDEO UNDERSTANDING

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ABSTRACT

Surgical video understanding is essential for intraoperative assistance such as workflow monitoring, instrument preparation and procedural documentation. Delivering such assistance in real time requires a model to jointly perceive the ongoing scene, remember past surgical events, and respond before clinically relevant moments pass. Existing surgical video-language models either operate offline on complete videos or narrate only the current frames, failing to retain procedural evidence and to decide when a response is required. In this work, we propose SurgStream, a workflow-anchored proactive streaming framework for real-time surgical video understanding. We first introduce Workflow-Anchored Memory (WAM) to record structured instrument–action–target observations and fold them into phase-level summaries, providing compact and queryable surgical memory. Deadline-aware Transition Learning (DTL) is then devised to supervise response states within clinically bounded windows, enabling timely responses before procedural deadlines. We further construct SurgStream-Bench, a benchmark comprising 11,667 streaming interactions over 14 tasks, five clinical scenarios and five temporal modes, where response timing is grounded in expert annotations. Extensive experiments demonstrate that SurgStream outperforms offline and streaming video large language models, particularly on memory and forward proactive tasks, while keeping the per-step update below the 1 s input interval.

***Index Terms*—** Surgical video understanding, streaming video LLM, workflow memory, proactive response

1. INTRODUCTION

Real-time surgical video understanding plays a crucial role in intraoperative assistance [1, 2], such as phase-aware instrument preparation, procedural documentation and progress tracking, where an assistant must perceive the ongoing scene, remember what has happened, and respond at the moment it is clinically needed. However, surgical videos are long, visually homogeneous and densely structured by the workflow [3], and a response is only useful before the next step begins, making real-time surgical interaction challenging.

Early surgical video-language models primarily focused on visual question answering on images or complete videos [4, 5, 6, 7]. Subsequently, general and medical vision-language models [8, 9, 10] achieved significant progress in visual understanding, and real-time surgical commentary [11] narrated the current frames on the fly. More recently, streaming video large language models (LLMs) [12, 13, 14, 15, 16] have enabled causal perception and proactive response on continuous video streams, and OneStreamer [17] further unifies perception, memory and proactive response in a single model.

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Despite the progress, these models still lack the surgical structure required for intraoperative interaction. On the one hand, generic streaming memory stores free-text captions whose boundaries are decided by the model itself, whereas surgical events are naturally delimited by phases and instrument–action–target triplets; such memory can hardly answer when an instrument first appeared or how many clips have been applied, and it grows linearly with surgery duration. On the other hand, generic timing learning only learns whether to respond, not the latest moment at which a response is still useful, while existing surgical evaluation [11] checks whether narration follows visual transitions rather than whether replies to queries and alerts arrive within clinically defined windows. Explicitly leveraging the surgical workflow offers a promising way to govern what to remember and when to respond.

To address these limitations, we propose SurgStream, a workflow-anchored proactive streaming framework for real-time surgical video understanding, which reshapes what a streaming video LLM writes into its memory and when it is taught to respond without adding network modules. Specifically, we propose a Workflow-Anchored Memory (WAM) that writes a timestamped *Observe* record whenever the instrument–action–target state changes stably and folds the records of each finished phase into a *Summary*, so that retrospective questions are answered from compact text that grows with the number of phases. We further devise a Deadline-aware Transition Learning (DTL) that derives a clinical window for every response target from expert labels and redraws the target response time inside it in each epoch, teaching the model to respond as early as the evidence allows and before the deadline. Moreover, we construct SurgStream-Bench, a streaming surgical interaction benchmark of 14 tasks across five clinical scenarios and five temporal modes (Fig. 1). Extensive experiments demonstrate that SurgStream outperforms offline and streaming video LLMs, particularly on memory and forward proactive tasks, with real-time efficiency on a single GPU.

2. METHODOLOGY

As shown in Fig. 2, SurgStream performs real-time surgical video interaction within a single streaming video LLM. Given a video stream $\{x_t\}_{t=1}^{T}$ arriving at 1 fps, the model reads at each step t the causal context $\mathcal{C}_t = [\mathcal{M}_t; \mathcal{V}_t; \mathcal{D}_t]$, consisting of the memory $\mathcal{M}_t$, the recent visual window $\mathcal{V}_t = \{x_{t-W+1}, \ldots, x_t\}$ and the dialogue $\mathcal{D}_t$, and emits a control state $s_t \in \{$*Silence*, *Standby*, *Response*, *Observe*, *Summary*$\}$ followed by optional text. On this basis, WAM and DTL anchor what to remember and when to respond to the surgical workflow without new network modules, and SurgStream-Bench evaluates real-time surgical video interaction.

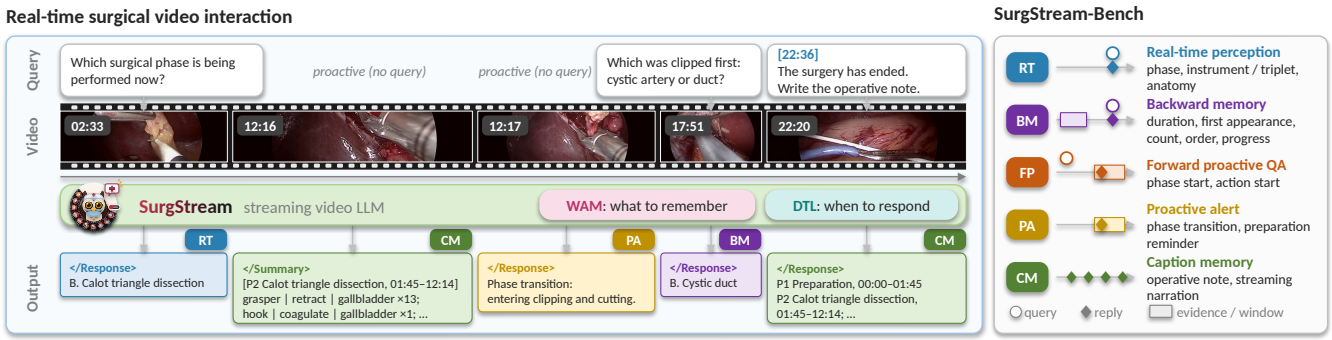


Fig. 1. Real-time surgical video interaction with SurgStream. Left: on a 1 fps test stream, SurgStream answers questions on current and past events, writes workflow-anchored memory and issues proactive alerts. Right: SurgStream-Bench covers five temporal modes over 14 tasks.

2.1. Workflow-Anchored Memory

To provide compact and queryable surgical memory, we propose WAM, which anchors both the write points and the content of memory to the surgical workflow instead of letting the model decide the boundaries of free-text captions. Let u_t denote the structured surgical state at time t, *i.e.*, the set of instrument–action–target triplets or present instruments. When u_t changes and remains stable for κ seconds, the model writes an *Observe* record $o_\tau = \langle t,\ u_t \backslash u_{t-1} \rangle$ at $\tau = t + \kappa - 1$, listing the newly appearing elements as "[mm:ss] instrument | verb | target". After a phase boundary b_{k+1} is confirmed, the model writes a *Summary* record $\sigma_k = \langle p_k, [b_k, b_{k+1}), \{(c, n_k(c))\} \rangle$ for the finished phase p_k, where $n_k(c)$ counts the onsets of element c in the *Observe* records $\mathcal{O}_k$ of this phase (*e.g.*, clip×3), and folds these records into it:

$$\mathcal{M}_{t+1} = (\mathcal{M}_t \setminus \mathcal{O}_k) \cup \{\sigma_k\}. \tag{1}$$

In this way, memory grows with the number of phases rather than with surgery duration, and questions on durations, first appearances and counts can be answered directly from the records. Since all records are derived from expert labels, memory writing is supervised by $\mathcal{L}_{\text{mem}}$, the token-averaged cross-entropy of each record given its state and causal context. WAM thus provides queryable surgical memory that is verifiable against expert labels and remains compact for long procedures.

2.2. Deadline-aware Transition Learning

To respond before clinically relevant moments pass, we propose DTL, which introduces the clinical window of each response target into state supervision, whereas generic sparse supervision [17] samples around a single annotated response time and ignores that a response remains acceptable until a deadline. Each response target j is assigned a window $[a_j, d_j]$, where the evidence time a_j and deadline d_j are derived from labels (*e.g.*, a preparation reminder is due at the next phase onset). As any response within the window is acceptable and earlier is better, we resample the ground-truth response time in each epoch:

$$r_j \sim p(r \mid a_j, d_j) \propto \exp\Big(-\frac{r - a_j}{\eta}\Big) \cdot \mathbb{I}[a_j \le r \le d_j], \tag{2}$$

where the temperature η controls the preference for early responses, and set $s^*_{r_j}$ =*Response*, s^*_t =*Standby* for $t \in [a_j - \epsilon, r_j)$, and *Silence* elsewhere. Following [17], the supervised set Ω comprises all output anchors (including the resampled r_j), state transitions and representative state-persistence positions, and the state loss is

$$\mathcal{L}_{\text{DTL}} = -\frac{1}{|\Omega|} \sum_{t \in \Omega} \log \pi_\theta(s^*_t \mid \mathcal{C}_t). \tag{3}$$

Keeping Ω sparse prevents the abundant *Standby* positions from dominating the loss and suppressing *Response*. As r_j is redrawn in each epoch with earlier times favoured by Eq. (2), DTL teaches the model not only whether but also when to respond, issuing replies as early as the evidence allows and before the deadline.

SurgStream is trained jointly with the cross-entropy of the reply text, $\mathcal{L}_{\text{mem}}$ and $\mathcal{L}_{\text{DTL}}$. At inference, the states are emitted in the order *Summary*→*Observe*→*Response* until *Silence* or *Standby*, and the context is bounded by the visual window and memory records, independent of surgery duration.

2.3. SurgStream-Bench

To evaluate real-time surgical video interaction, we construct SurgStream-Bench from two public laparoscopic cholecystectomy datasets: Cholec80 [3] (80 videos with 7 phases and 7 tools) and CholecT50 [23] (50 videos with phases and 100 triplet classes). All videos are used at 1 fps. Since the two datasets share recordings, we assign splits per procedure following the official overlap mapping and dataset splits [24], and load each shared recording once with both label sets merged. This yields 53/11/21 procedures (32.5/6.1/15.5 h) for training/validation/test with no test procedure seen in training or validation. Following OneStreamer-1M [17], the 14 tasks cover five temporal modes, *i.e.*, real-time perception (RT), backward memory (BM), forward proactive QA (FP), proactive alert (PA) and caption memory (CM), across five clinical scenarios: workflow (S1), instrument and action (S2), anatomy (S3), documentation (S4) and narration (S5). All answers, evidence times and deadlines are derived from expert labels, *e.g.*, a phase-transition alert must be issued within $[b_k, b_k + 10\,\text{s}]$ and a preparation reminder within $[b_k - 60\,\text{s}, b_k]$. In total, SurgStream-Bench contains 11,667 instances (7,315/1,320/3,032), of which 1,893 require proactive timing (FP/PA).

Questions on the current phase, instruments or triplets, anatomy and the order of two handled structures are multiple-choice and scored by accuracy (Acc); durations, first-appearance times and counts are correct within 20%, 10 s and ±1, respectively; and the progress question (what has been done so far) is scored by S-F1. For FP, timing accuracy (T-Acc) counts the first reply after the query as correct only if it falls in $[a_j, d_j]$ with correct content. For PA, replies are matched one-to-one in time to ground-truth events, and timing F1 (T-F1) counts a hit only inside $[a_j, d_j]$ with correct content. For CM, structured F1 (S-F1) averages the F1 of phase intervals and triplet events, with boundaries and event times matched within 30 s and narration due within 10 s after each phase. Content is checked by exact vocabulary matching, so no LLM judge is needed. Each mode or scenario score averages its tasks, and Avg_M and Avg_S average the

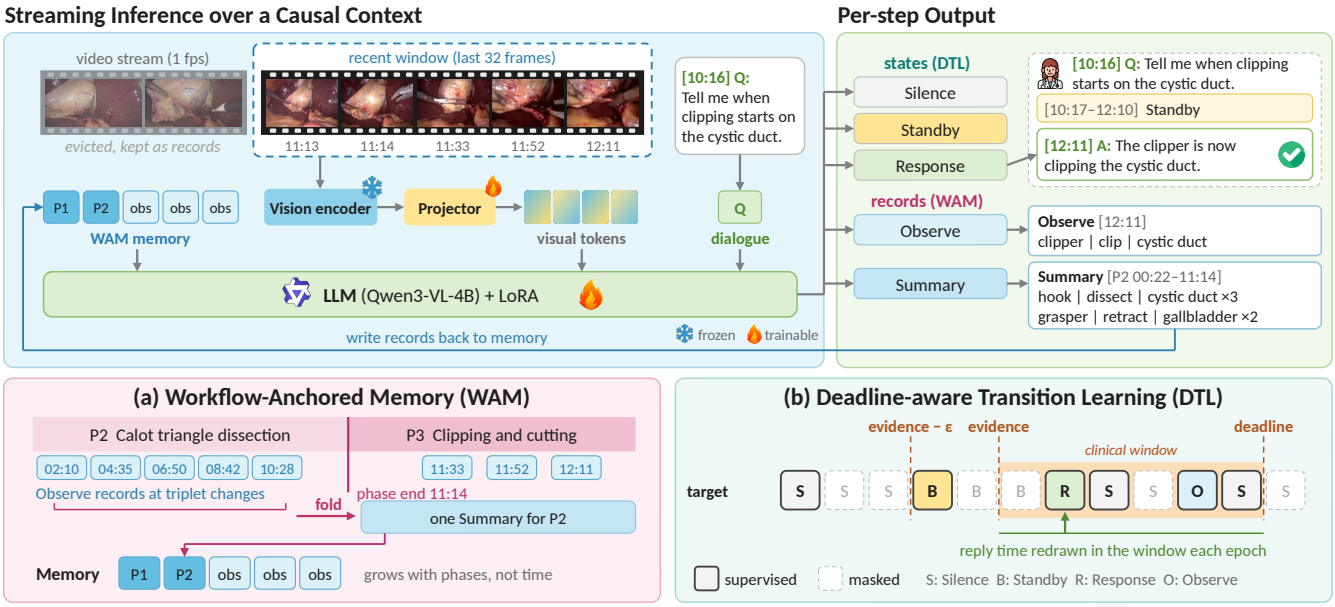


Fig. 2. Overview of SurgStream. At each step, the LLM reads WAM records, the recent visual window and pending dialogue, and emits a control state with optional text. (a) WAM writes *Observe* records at triplet changes and folds them into a *Summary* at each phase boundary. (b) DTL redraws the response time within the clinical window in each epoch and supervises states on a sparse mask.

Table 1. Comparison with state-of-the-art methods on SurgStream-Bench. †: fine-tuned on the SurgStream-Bench training split. The best and second-best results are in **bold** and underlined, respectively. All scores are reported in %.

Methods	Size	RT Acc↑	BM Score↑	FP T-Acc↑	PA T-F1↑	CM S-F1↑	Avg_M ↑	S1	S2	S3	S4	S5	Avg_S ↑
Qwen3-VL [8]	4B	41.73	29.93	0.43	0.30	4.26	15.33	12.11	30.13	46.95	0.68	8.52	19.68
Qwen3.5 [18]	4B	42.70	27.77	2.69	1.08	5.98	16.04	13.06	28.88	43.46	5.38	7.93	19.74
Gemma 4 [19]	E4B	42.89	19.88	0.86	1.60	6.62	14.37	14.17	13.06	52.39	4.64	9.08	18.67
MedGemma [20]	4B	34.54	11.14	0.43	0.73	3.48	10.06	8.84	11.04	36.66	0.89	5.17	12.52
Hulu-Med [10]	4B	42.77	29.70	0.00	0.78	5.95	15.84	13.18	30.15	42.23	6.01	7.28	19.77
MedVLM [21]	7B	55.35	27.80	0.54	1.84	2.43	17.59	13.75	33.36	53.13	0.68	4.85	21.16
EndoChat [22]	13B	39.62	27.92	0.00	0.21	0.86	13.72	5.29	29.57	55.37	2.33	0.48	18.61
OneStreamer† [17]	4B	83.27	59.10	10.87	14.78	15.16	36.64	32.99	62.72	74.81	18.49	24.47	42.70
SurgStream	4B	**84.07**	**64.13**	**23.13**	**16.64**	**25.22**	**42.64**	**36.84**	**67.70**	**78.06**	**31.45**	**28.79**	**48.57**

Table 2. Ablation study of SurgStream on SurgStream-Bench. The first row is the baseline without WAM and DTL. The best results are in **bold**. All scores are reported in %.

WAM	DTL	RT↑	BM↑	FP↑	PA↑	CM↑	Avg_M ↑	Avg_S ↑
		83.27	59.10	10.87	14.78	15.16	36.64	42.70
✓		82.74	62.46	14.53	**17.58**	22.84	40.03	46.74
	✓	83.48	61.97	20.23	14.37	22.34	40.48	45.96
✓	✓	**84.07**	**64.13**	**23.13**	16.64	**25.22**	**42.64**	**48.57**

five modes and the five scenarios, respectively.

3. EXPERIMENTS

3.1. Implementation Details

We initialize SurgStream from OneStreamer-4B [17] built on Qwen3-VL-4B [8] and adopt two-stage LoRA fine-tuning: Stage 1 warms up memory writing for 1 epoch, and Stage 2 jointly trains all 14 tasks for 2 epochs. As proactive targets and phase boundaries are rare, two extra training windows are centred on each FP/PA instance and each *Summary* write point. To mimic self-generated memory, the ground-truth memory prefix is perturbed by record dropping ($p = 0.1$), timestamp jittering (± 2 s) and random element replacement ($p = 0.05$). We use $W = 32$, $\kappa = 3$ s, $\epsilon = 5$ s, $\eta = 10$ s, and apply LoRA with rank 16 and AdamW with an initial learning rate of 1×10^{-4}, cosine annealing and a global batch size of 32. States and text are decoded greedily. For fair comparison, OneStreamer† uses the same data, window sampling and epochs as SurgStream. Offline VLMs are evaluated in a pseudo-streaming manner by uniformly sampling up to 32 frames from the video start to the query and polling every 5 s with the last 8 frames for FP/PA. Experiments run on RTX 6000 Ada and RTX 5090 GPUs.

3.2. Comparison with State-of-the-Art Methods

We compare SurgStream with general offline VLMs (Qwen3-VL [8], Qwen3.5 [18], Gemma 4 [19]) and medical VLMs (MedGemma [20], Hulu-Med [10], MedVLM [21], EndoChat [22]) in the zero-shot setting, and with the streaming video LLM OneStreamer [17] fine-tuned on the same data (OneStreamer†). As shown in Table 1, SurgStream achieves the best Avg_M of 42.64%,

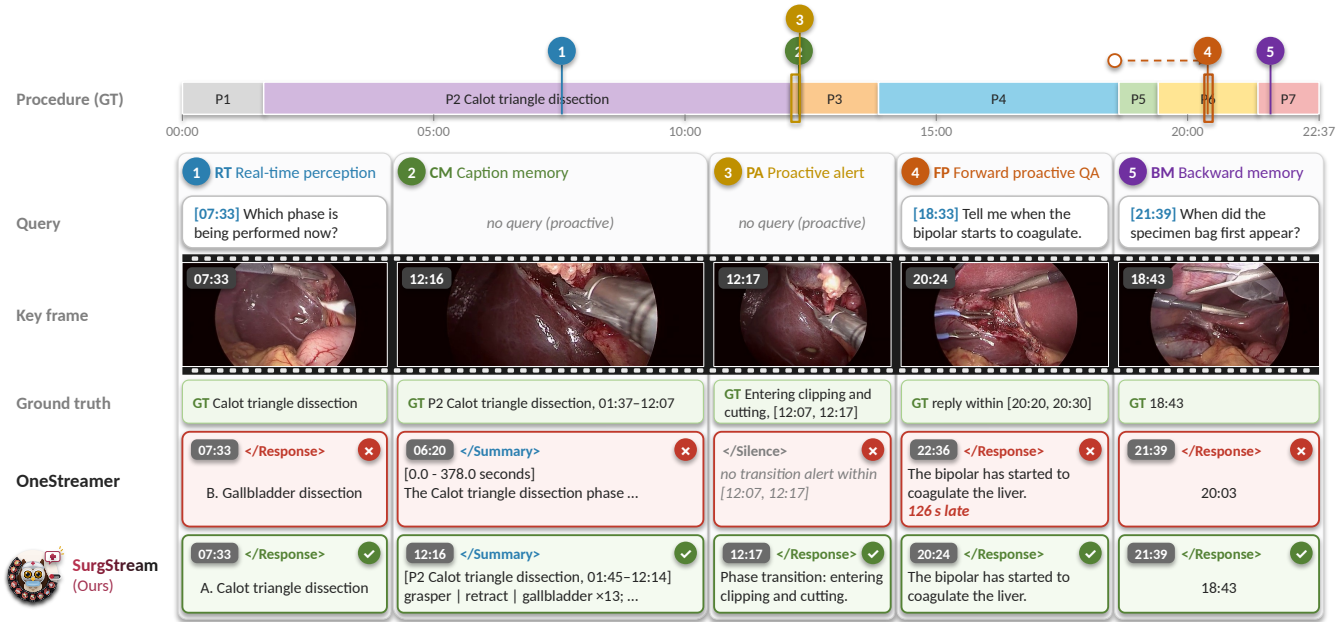


Fig. 3. Qualitative comparison of OneStreamer and SurgStream on one test procedure across the five temporal modes. SurgStream answers correctly in all five examples, recalling past events from its workflow-anchored memory and issuing proactive replies within the clinical windows, whereas OneStreamer gives wrong, late or missing replies in these examples.

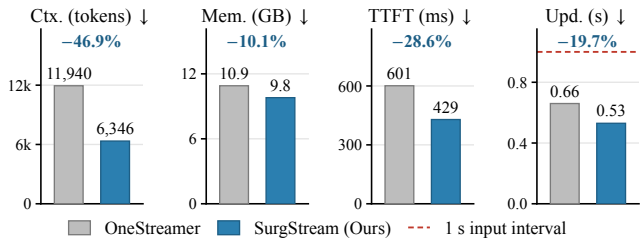


Fig. 4. Efficiency comparison with OneStreamer on full-length test videos at 1 fps on a single GPU. SurgStream reduces all four costs and keeps the per-step update below the 1 s input interval.

surpassing the second-best OneStreamer† by 6.00% (48.57% *vs.* 42.70% on Avg_S), with the largest gains on FP (+12.26%) and CM (+10.06%), highlighting the benefit of workflow-anchored memory and deadline-aware transition learning. Zero-shot VLMs stay below 3% on FP and PA, since offline models answer only when queried and cannot decide when to respond, and medical VLMs do not consistently outperform general ones (*e.g.*, MedGemma 10.06% *vs.* Qwen3-VL 15.33%). Across scenarios, SurgStream improves all five and gains most on documentation (S4, +12.96%), which relies on phase-level summaries. In the example of Fig. 3, OneStreamer misidentifies the current phase, misses the phase-transition alert and replies to the coagulation query 126 s after its deadline, whereas SurgStream answers all five correctly and on time.

3.3. Efficiency Analysis

We measure context tokens (Ctx.), GPU memory (Mem.), time to first token (TTFT) and per-step update time (Upd.) on full-length test videos. As shown in Fig. 4, SurgStream reduces context tokens by 46.9% and keeps the per-step update time of 0.53 s below the 1 s input interval, enabling real-time interaction. The saving comes from WAM: structured records replace free-text captions, and the memory grows with the number of phases rather than with surgery duration. Compared with OneStreamer, the shorter context also lowers TTFT by 28.6% and GPU memory by 10.1%, so that responses start sooner after each new frame.

3.4. Ablation Study

We ablate both components on top of a baseline that uses free-caption memory and sparse state supervision (PSTL) [17]. This baseline is identical to OneStreamer† in Table 1, so all variants share the same data and training schedule. As shown in Table 2, introducing WAM improves BM by 3.36% and CM by 7.68%, DTL improves FP by 9.36% and CM by 7.18%, and combining both achieves the best Avg_M of 42.64% and Avg_S of 48.57%. Both components improve CM by over 7%; WAM brings the larger gain on BM and PA, whereas DTL brings the larger gain on FP. The combination performs best on all modes except PA, where WAM alone is higher (17.58% *vs.* 16.64%). Still, FP and PA stay below 24% for all variants, so response timing remains the main difficulty.

4. CONCLUSION

We presented SurgStream, a proactive streaming framework that uses the surgical workflow to govern what to remember (WAM) and when to respond (DTL). WAM anchors memory writing to phases and triplet changes, so that the memory stays queryable and grows with the number of phases rather than with surgery duration. DTL supervises response states within clinically bounded windows, teaching the model to respond as soon as the evidence allows and before the deadline. On SurgStream-Bench, where response timing is grounded in expert annotations, SurgStream outperforms offline and streaming video LLMs, particularly on memory and proactive tasks, while keeping the per-step update below the 1 s input interval. Nevertheless, proactive tasks remain far from solved, and the benchmark currently covers a single procedure. Future work will extend SurgStream to other surgical procedures and to safety assessment such as the critical view of safety.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by Cholec80 [3] and CholecT50 [23]. Ethical approval was not required as confirmed by the licenses attached with the open access data.

6. ACKNOWLEDGEMENTS

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